Assignment-2

Writing Guidelines: `

1. Use clean A4 sheets for your assignments, ensuring legible handwriting.

2. Include the assignment number, course with section number, your name, and student ID on the first/cover page.

3. Number each page for easy reference.

4. Write your student ID on the top right corner of every assignment page.

5. Clearly label all your answers with the correct corresponding numbers.

1. A researcher aims to investigate the relationship between body mass index (BMI) and blood pressure levels. To do so, the researcher collects data from a sample of 25 participants from a particular area, recording each participant's BMI and their corresponding systolic blood pressure readings. **[4+2+1+5+4]**

Participant	BMI	Blood Pressure (mmHg)
1	22.5	120
2	27.1	130
3	30.3	140
4	24.2	125
5	29.1	135
6	26.9	130+**R**
7	31.5	145
8	23.8	122
9	28.4	138
10	32.0	150
11	21.6	120-**R**
12	25.6	126
13	33.3	155
14	20.7	115
15	27.9	136-**R**
16	30.0	139
17	26.2	127
18	29.8	137
19	22.1	119
20	34.5	158
21	28.0	135+**R**
22	31.1	140+**R**
23	24.7	124
24	32.8	152
25	23.4	121

[R = ***last digit of your student ID***]

a) Fit a linear regression model of Blood Pressure on BMI.

b) Provide interpretation of regression parameters.

c) Using parameter value from predict the Blood Pressure if the BMI is 22.5.

d) Find the Error Sum of Square

e) Does your model predict well? Provide a mathematical explanation.

2. Consider the experiment of tossing an unbiased dice and a coin once simultaneously. Let A be the event of getting 5 and B be the event of getting Head. **[1+2+3+3]**

    a) Write the sample space of this experiment.

    b) If you get event A, what is probability of getting event B?

    c) Are events A and B independent or not and why?

    d) If you do not get 6, what is the probability that you will get Head.